

eter, the correct number of outliers was used. For the BCP algorithm, a range of thresholds on the posterior probability of change-points was used to detect a range of number of segments. As is evident from the empirical results below, the ORCS algorithm can achieve high detection rate of the outliers even without knowing their exact number a-priori. Furthermore, we suggest below a way of estimating the number of outliers. For each algorithm, the maximal R measure over all parameters range was used to compare all algorithms.

Results Results are shown in Fig. 3(b) as the maximal R measure achieved versus the percentage of outliers, for each of the algorithms considered. It is evident that the performance of the BU and BCP algorithms decreases significantly as more outliers are added, while the outlier-robust ORCS algorithm keeps an approximately steady performance. Our unweighted and weighted TD-ORCS algorithms achieve the best performance for all levels of outliers. Results for LARS algorithm are omitted as it did not perform better than other algorithms. We verified the ability of our algorithms to correctly detect outliers by calculating the R measure of the outliers detection of the ORCS algorithm, with zero length tolerance window, i.e a detection is considered a true-positive only if it exactly pinpoints an outlier. The R measure of the detection was evaluated on the γ , λ parameter grid, as well as the corresponding numbers of detected outliers. Results for the representative case of $p = 10\%$ outliers are shown in Fig. 4(a). It is evident that a high R measure (> 0.9) is attained on a range of parameters that yield around the true number of outliers. We conclude that one does not need to know the exact number of outliers in order to use the ORCS algorithm, and a rough estimate can be approximated from the histogram of number of detected outliers (i.e. Fig. 4(b)).

4 Related Work and Conclusion

There is a large amount of literature on change-point detection, see for example (?; ?). Optimal segmentation can be found using dynamic programming (?); however, the complexity of this approach is quadratic in the number of samples n , and therefore might be infeasible for large data sets. Some approaches which achieve complexity linear in n (?; ?) treat only one dimensional data. Some related work is concerned with the objective Eq. (3) we presented in Sec. 2.1. In (?) it was suggested to reformulate Eq. (3) for the one dimensional case as a LASSO regression problem (?; ?), while (?) extended this approach to multidimensional data, although not treating outliers directly. Another common approach is deriving an objective from a maximum likelihood criterion of a generative model, and then either optimize the objective or use it as a criterion for a top-down or a bottom-up approach (?; ?; ?; ?). We note that the two-dimensional version of Eq. (3) is used in image denoising applications, where it is known as the Total-Variation of the image (?; ?; ?). Finally, we note that all these approaches do not directly incorporate outliers into the model.

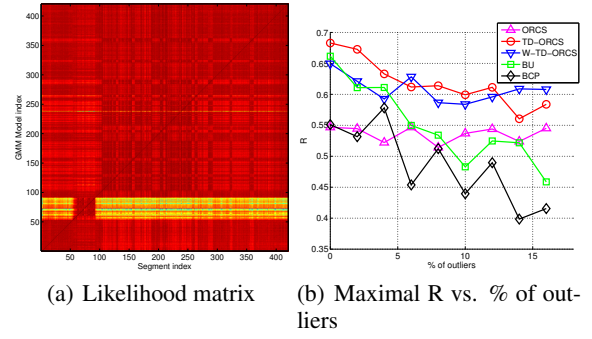


Figure 3: (a) A_{ij} is the log-probability of segment j given the GMM fitted to segment i . (b) Maximal R measure vs. percentage of outliers. See text for algorithms details.

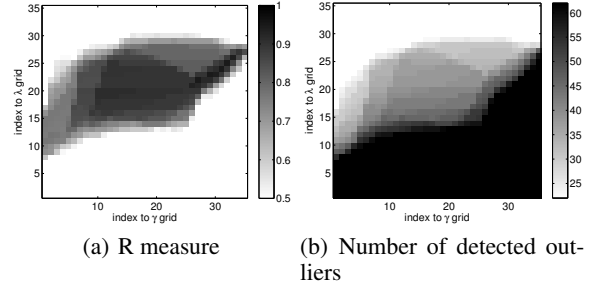


Figure 4: (a) Outliers detection quality by the ORCS algorithm, quantified using the R measure of outliers detection. R is evaluated on a 35×35 parameter grid, for 10% outliers. (b) The corresponding number of detected outliers.

We formulated the task of segmenting sequential data and detecting outliers using convex optimization, which can be solved in an alternating manner. We showed that a specific choice of weighting can empirically enhance performance. We described how to calculate λ^* and γ^* , the critical values for the split into two segments and the detection of the first outlier, respectively. These values are useful for finding a solution path in the two-dimensional parameter space. We also derived a top-down, outlier-robust hierarchical segmentation algorithm which minimizes the objective in a greedy manner. This algorithm allows for directly controlling both the number of desired segments K and number of outliers M . Experiments with real-world audio data with outliers added manually demonstrated the superiority of our algorithms.

We consider a few possible extensions to the current work. One is deriving algorithms that will work on-the-fly. Another direction is to investigate more involved noise models, such as noise which corrupts a single feature along all samples, or a consecutive set of samples. Yet another interesting question is how to identify that different segments come from the same source, e.g. that the same speaker is present at different locations in a recording. We plan to investigate these directions in future work.

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